Olle Marmenlind

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Final year Computer Engineering student at KTH while working as Technical Product Owner at Sevan AB, will transition into systems engineering late 2026.
Has internship experience with low-level and hardware aware software, as well as data ingestion pipelines, distributed systems, and observability.
My current focus is to deepen expertise in areas i enjoy and challenge me the most: backend infrastructure and network infrastructure, runtimes and compilers, embedded systems, and performance oriented platform development.

Education

Royal Institute of Technology (KTH) — Stockholm, Sweden
B.Sc. in Computer Engineering — **Expected June 2027**

- **Relevant Coursework:** Data Structures & Algorithms, Distributed Systems, Operating Systems, Embedded Systems/Microcontrollers (graduate level), Computer Networks, Databases, Network Security (graduate level)
- **Certifications:** CCNA, CCNP

Experience

Sevan AB — Stockholm, Sweden (Remote)
Technical Product Owner — **March 2022 – Present**

- Promoted to Product Owner; scaled B2C content platform from 4.5K to 200K users and migrated 300+ B2B accounts to a digital e-commerce channel.
- Directed product roadmap and authored 400+ Jira tickets including API schemas and data models.
- Collaborated cross-functionally to translate business requirements into technical solutions and development priorities, coordinated development across external partner teams and vendors.

Motillo AB — Stockholm, Sweden
Software Engineer Intern — **March 2026 – June 2026**

- Built the React/TypeScript SPA (Entra ID auth) as a two-way control plane for UptimeRobot, managing tenant monitors, and for Litium order ingestion, built specifically to run continuously on legacy Chrome 76 kiosk devices.
- Implemented a simulated Lambda Architecture using PostgreSQL Materialized Views for historical rollups, merging them with live data to drop query times from 330ms to ∼20ms over 4 million rows.
- Set up an asynchronous ingestion buffer using Hangfire and Polly to handle upstream API rate limits and prevent frontend load delays. Implemented observability layer with OpenTelemetry.
- Wrote a C#/ASP.NET Core Web API extension for the Litium platform to extract, transform, and serve flattened Elasticsearch order data.

Proximion AB — Stockholm, Sweden
R&D Systems Engineer Intern — **October 2025 – January 2026**

- Built a multi-threaded processing pipeline (FreeRTOS, C), using non-blocking IPC queues to synchronize high-frequency data ingestion with asynchronous I/O tasks.
- Designed a custom binary transport protocol featuring a deterministic state machine for byte-framing, sequence tracking, and payload CRC validation across hardware boundaries (MCU ↔ PC host).
- Wrote kernel-level memory hooks to trap stack overflows and heap exhaustion, using out-of-band hardware signaling to extract thread-specific crash states in a debugger-less environment.
- Configured a cross-platform GNU Make build system and self-hosted Linux CI/CD pipeline utilizing hermetic Docker containers to enforce reproducible builds, automated format patching, and release generation.

Skills

- **Languages:** Java, C++, Go, C, C#, TypeScript
- **Systems & Infrastructure:** Linux, FreeRTOS, RISC-V, Docker, CI/CD, Microservices, Distributed Systems
- **Backend:** PostgreSQL, Elasticsearch, ASP.NET Core, Spring Boot, REST, ETL/data pipelines, background jobs
- **Tools & Networking:** Git, Bash, PowerShell, GNU Make, TCP/UDP, OpenTelemetry